

Installing Beliefs to Audit Attribution: A Ground-Truth Testbed and Five Ways It Misleads

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1 Motivation: measured causal-effect testbed and validation target

The testbed is motivated as a validation target grounded in measured causal effect sizes, rather than as a first-of-its-kind dataset. This is contribution framing, not an empirical result.

2 Methodology: estimands, controls, scope, and reporting rules

Absorption is operationalized as held-out premise specialization rather than a belief measure. Belief and action contrasts are netted against their matched off-topic controls and are not interpreted as evidence of causal mediation. Raw belief and action checkpoint figures serve as diagnostics rather than netted transfer estimates.

The analysis is exploratory, with model, topic, seed count, and checkpoint scope limited to recorded artifacts. Two-seed scope applies only to declared seed-paired contrast families, while single-seed and instrument-specific readings are identified separately; the evidence bundle is therefore not treated as uniformly seed-paired.

Fixed tables are authoritative for numerical estimates and confidence intervals rather than prose.

3 Results: instrument demonstration and five validation failures

The belief instrument was demonstrated by an explicit-stance positive control at the earlier single-seed, instrument-specific reading. This point estimate was derived from recorded arm scores without an inferred interval; its token dose was lower than that of the evidence corpus, and it does not establish general belief transfer. At the later reading, the same positive-control contrast decayed while the matched-control-netted evidence-only contrast rose. The earlier evidence-only reading was positive despite being quoted in earlier drafts as the near-null cell; it used a matched off-topic control, all arms gated, and a single seed. With the same arms and control later, the evidence-only contrast was larger, supporting a slow rather than inert interpretation. All seven arms gated, and because the control passed at the recorded threshold, the late-step gate failure was genuinely late. The deterministic table pairs each belief ratio with its optimizer-step reading.

The form cross exposed instability rather than a uniform form effect. Across three seeds, the short, dense form produced the strongest recorded form-cross belief contrast and was the only band-level finding; it used the same premise specification as the long-form arms, leaving form as the only variable. At matched density, the long form yielded a smaller positive contrast. The short, sparse corner remained directional but unstable across three seeds and carries no ratio.

The model-size cross separated belief from action. The matched-control-netted explicit-stance belief contrast was positive for both models: the smaller-model arm was dose-matched to the larger-model arm with model as the only difference, while the larger-model control was retrained at its own size and used its own calibrated gate threshold rather than the smaller-model bar. The smaller-model matched-control-netted action contrast was unresolved because its interval straddled zero, so no conduction ratio is quotable. The larger-model action contrast was positive and replicated in direction at the paired seed, but its magnitude is not claimed because it scattered across seeds. These action results do not establish causal mediation.

Full fine-tuning yielded positive, matched-control-netted evidence-corpus belief contrasts in both the first declared seed reading and the seed-paired replication. The paired reading used a seed-matched full-fine-tuning control and a re-derived schedule. Under LoRA, the same corpus straddled zero on probability while excluding zero on log-odds, making this scale dependence rather than a capacity cliff.

The control-machinery term—the quantity removed by netting—varied across each declared LoRA seed pair. For both the smaller- and larger-model LoRA families, the paired intervals did not overlap; the larger-model readings also had opposite signs. By contrast, the full-fine-tuning control-machinery term held comparatively still across its declared seed pair, with overlapping intervals, unlike every recorded LoRA control family.

The software-architecture second-topic validation did not produce a stable netted sign across three declared seed readings. The first reading was near zero with a matched off-topic control, all arms gated, and was one of the three sign-disagreeing readings. The second was also near zero with a seed-matched control. The third reversed sign despite all five arms gating and a seed-matched control; the sign disagreement arose in the control’s machinery term rather than the treatment contrast.

The descriptive-inference axis was withdrawn because its explicit-arm positive control was inverted and could not demonstrate premise-specificity. This is an instrument-failure result rather than a caveat on an otherwise usable transfer estimate: the byte-identical null-control facet was the highest-moving of all eight facets, and no transfer number from the suite is reported. The single-seed, instrument-specific point estimate was derived from recorded arm scores, and no interval was inferred.

Table 1: Declared contrasts selected or derived from immutable result summaries. Source: `paper/synthesis.json`.

intervention	net effect	qualification
Second topic, evidence-only, belief, seed 42	+0.0076 [-0.0008, +0.0159]	Second topic (software architecture), matched off-topic control, all arms gated. Reported as one of three seeds whose netted signs disagree.
Second topic, evidence-only, belief, seed 7	+0.0122 [+0.0053, +0.0202]	Seed-seven replicate with a seed-matched control.
Second topic, evidence-only, belief, seed 123	-0.0034 [-0.0156, +0.0089]	Third seed with a seed-matched control, all five arms gated. Its sign disagrees with the other two seeds; the disagreement arises in the control’s machinery term, not the treatment contrast.
Control machinery, 4B LoRA, seed 42	-0.0173 [-0.0250, -0.0106]	The quantity netting subtracts. Compare against the seed-7 reading of the same family.
Control machinery, 4B LoRA, seed 7	-0.0025 [-0.0078, +0.0036]	Its interval does not overlap the seed-42 reading of the same family.
Control machinery, 8B LoRA, seed 42	-0.0031 [-0.0069, +0.0008]	Compare against the seed-7 reading, whose sign differs.
Control machinery, 8B LoRA, seed 7	+0.0156 [+0.0086, +0.0226]	Its interval does not overlap the seed-42 reading, and the two carry opposite signs.
Control machinery, full fine-tuning, seed 42	-0.0053 [-0.0101, -0.0014]	The full fine-tuning control, whose two seed intervals overlap.
Control machinery, full fine-tuning, seed 7	-0.0049 [-0.0090, -0.0012]	Overlaps its seed-42 counterpart, unlike every LoRA control family.
Evidence-only, belief, step 24 (2 epochs)	+0.0072 [+0.0016, +0.0136]	The reading the earlier drafts quoted as the near-null cell. Matched off-topic control, all arms gated.
Evidence-only, belief, step 36 (3 epochs)	+0.0342 [+0.0248, +0.0443]	Same arms, same control, three epochs. All seven arms gated; the control passes at 0.802, so the late-step gate failure is genuinely late. One seed.
Explicit stance, belief, step 24	+0.3111	Positive control for the belief instrument. Token dose is lower than the evidence corpus. Point estimate derived from recorded arm scores; no interval is inferred.
Explicit stance, belief, step 36	+0.2167	The same positive control one epoch later, where it has begun to decay while the evidence cell is still rising. Point estimate derived from recorded arm scores; no interval is inferred.
Short, dense (Ms3p), belief	+0.1190 [+0.0901, +0.1497]	Same premise specification as the long-form arms; form is the only variable. Band-level across three seeds.
Long, dense (Mld), belief	+0.0547 [+0.0391, +0.0705]	The length arm of the form cross at matched density.
Short, sparse (Mss), belief	+0.0506 [+0.0308, +0.0720]	The unstable corner: scatters 2.98x across three seeds, so its reading is directional and carries no ratio.
Explicit stance at 4B, belief	+0.1517 [+0.1109, +0.1942]	Dose-matched to the 8B arm; model is the only difference.
Explicit stance at 8B, belief	+0.0970 [+0.0826, +0.1115]	Control retrained at 8B. Qwen3-8B has its own calibrated gate threshold; the 4B bar is not reused.
Explicit stance at 4B, action	+0.0011 [-0.0101, +0.0121]	Straddles zero. No conduction ratio is quotable while this numerator straddles.
Explicit stance at 8B, action	+0.0352 [+0.0249, +0.0459]	Excludes zero. Replicated in direction at seed 7; scatters 44 percent across seeds, so no magnitude.
Full fine-tuning, evidence corpus, belief, seed 42	+0.0164 [+0.0060, +0.0282]	Under LoRA the same corpus straddles zero on probability and excludes zero on log-odds, so the contrast is scale-dependent rather than a capacity cliff.
Full fine-tuning, evidence corpus, belief, seed 7	+0.0115 [+0.0046, +0.0199]	Seed-seven replication with a seed-matched full-FT control and a re-derived schedule.
Descriptive-inference suite, explicit arm (the INVERTED positive control)	+0.0620	This entry is the instrument-failure result. On this arm the byte-identical null-control facet is the highest-moving of all eight facets, so the suite cannot demonstrate premise-specificity and no transfer number from it is reported. Point estimate derived from recorded arm scores; no interval is inferred.

Note. Rows without intervals are deterministic point contrasts over recorded arm scores.

Table 2: Recorded belief contrast and sensitivity quantities. Source: matrix_v1_step24/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0033 [-0.0009, +0.0082]
machinery	-0.0039 [-0.0093, +0.0007]
delta_net	+0.0072 [+0.0016, +0.0136]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0110

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 3: Recorded belief contrast and sensitivity quantities. Source: matrix_v1_step36/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0228 [+0.0154, +0.0307]
machinery	-0.0114 [-0.0175, -0.0050]
delta_net	+0.0342 [+0.0248, +0.0443]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0524

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 4: Recorded belief contrast and sensitivity quantities. Source: h8_4b/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.1344 [+0.0930, +0.1779]
machinery	-0.0173 [-0.0250, -0.0106]
delta_net	+0.1517 [+0.1109, +0.1942]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 5: Recorded action contrast and sensitivity quantities. Source: h8_8b/action_summary.yaml.

quantity	recorded value
delta_raw	+0.0201 [+0.0128, +0.0280]
machinery	-0.0151 [-0.0215, -0.0090]
delta_net	+0.0352 [+0.0249, +0.0459]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

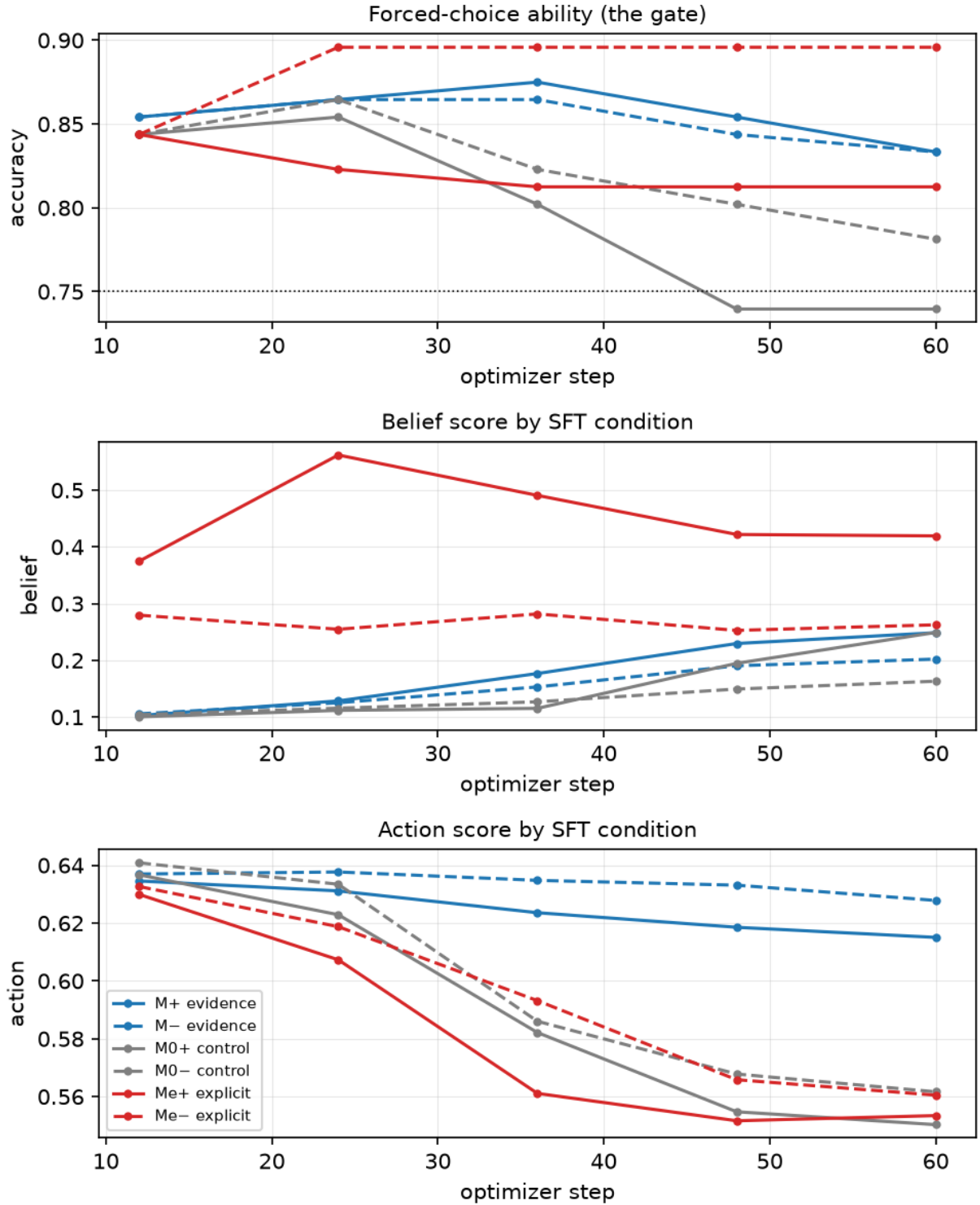


Figure 1: Render the declared matrix run's saved figures; label raw belief or action curves as diagnostic; direct all numerical estimates and intervals to fixed tables..

4 Discussion: detection recipes, withdrawal, and bounded generality

The evidence-only belief contrast is positive at the earlier recorded checkpoint, although earlier drafts treated this as the near-null cell. This single-seed reading uses a matched off-topic control and has all arms gated; the deterministic table pairs the belief ratio with its optimizer-step reading. With the same arms and control, the later single-seed contrast is larger, supporting a slow rather than inert interpretation. All seven arms are gated, and the control passes at the recorded threshold, so the gate failure is genuinely late; the deterministic table again pairs the belief ratio with its optimizer-step reading.

The descriptive-inference axis is withdrawn because its explicit-arm positive control is inverted and therefore cannot demonstrate premise-specificity. This is an instrument-failure result rather than a caveat on an otherwise usable transfer estimate: the byte-identical null-control facet is the highest-moving of all eight facets, and no transfer number from this suite is reported. The reading is instrument-specific and single-seed, its point estimate is derived from recorded arm scores, and no interval is inferred.

The smaller-model LoRA control-machinery term varies across the declared seed pair, with paired intervals that do not overlap; this is the quantity netting subtracts. The larger-model LoRA term also varies across the declared seed pair, with non-overlapping paired intervals and opposite signs; this likewise is the quantity netting subtracts. By contrast, the full-fine-tuning control-machinery term holds comparatively still across the declared seed pair: its paired intervals overlap, unlike every recorded LoRA control family.

The model evidence supports replication across the two tested model sizes and configurations, rather than broader model-axis generality, because it establishes only positive explicit-stance belief contrasts under different model-specific gates. Topic generality is not supported: the second topic’s declared seed contrasts disagree in sign. This bounded conclusion is a synthesis of the atomic model and second-topic results.

5 Appendix: remaining run-level deterministic tables

The remaining run-level form tables record the strongest form-cross belief contrast for the short, dense form, the smaller positive contrast for the long, dense length arm at matched density, and the directional but unstable short, sparse corner. The short, dense comparison spans three seeds and holds the premise specification fixed relative to the long-form arms, leaving form as the only variable; it is the paper’s only band-level finding. The short, sparse reading is unstable across three seeds and carries no ratio.

The remaining model and capacity tables record a positive matched-control-netted explicit-stance belief contrast for the larger model, whose control was retrained at that size and whose calibrated gate threshold replaces rather than reuses the smaller-model bar. The smaller-model matched-control-netted action contrast remains unresolved because its interval straddles zero, leaving no quotable conduction ratio and no basis for causal-mediation claims. Full fine-tuning records positive matched-control-netted evidence-corpus belief contrasts in the first declared seed reading and the seed-paired replication; the latter uses a seed-matched full-fine-tuning control and a re-derived schedule. For the same corpus under LoRA, probability straddles zero while log-odds excludes zero, indicating scale dependence rather than a capacity cliff.

The remaining software-architecture tables record near-zero second-topic belief contrasts in the first two declared seed readings and a sign reversal in the third. The first uses a matched off-topic control with all arms gated; the second uses a seed-matched control. Both belong to the three readings whose netted signs disagree. The third also uses a seed-matched control and has all five arms gated, with the sign disagreement arising in the control’s machinery term rather than the treatment contrast.

A Supplementary diagnostics

Table 6: Recorded belief contrast and sensitivity quantities. Source: h8_8b/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0938 [+0.0787, +0.1091]
machinery	-0.0031 [-0.0069, +0.0008]
delta_net	+0.0970 [+0.0826, +0.1115]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 7: Recorded action contrast and sensitivity quantities. Source: h8_4b/action_summary.yaml.

quantity	recorded value
delta_raw	+0.0013 [-0.0083, +0.0100]
machinery	+0.0001 [-0.0049, +0.0049]
delta_net	+0.0011 [-0.0101, +0.0121]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 8: Recorded belief contrast and sensitivity quantities. Source: h19_full_ft/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0111 [+0.0030, +0.0212]
machinery	-0.0053 [-0.0101, -0.0014]
delta_net	+0.0164 [+0.0060, +0.0282]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0251

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 9: Recorded belief contrast and sensitivity quantities. Source: h19_full_ft_s7/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0066 [+0.0012, +0.0137]
machinery	-0.0049 [-0.0090, -0.0012]
delta_net	+0.0115 [+0.0046, +0.0199]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0176

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 10: Recorded belief contrast and sensitivity quantities. Source: ms3p_arms/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.1034 [+0.0741, +0.1352]
machinery	-0.0156 [-0.0271, -0.0048]
delta_net	+0.1190 [+0.0901, +0.1497]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.1825

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 11: Recorded belief contrast and sensitivity quantities. Source: mld_arms/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0392 [+0.0279, +0.0514]
machinery	-0.0156 [-0.0271, -0.0048]
delta_net	+0.0547 [+0.0391, +0.0705]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0839

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 12: Recorded belief contrast and sensitivity quantities. Source: ms_sparse_arms/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0350 [+0.0237, +0.0472]
machinery	-0.0156 [-0.0271, -0.0048]
delta_net	+0.0506 [+0.0308, +0.0720]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0776

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 13: Recorded belief contrast and sensitivity quantities. Source: software_architecture/sw_arms_v1/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0155 [+0.0084, +0.0233]
machinery	+0.0079 [+0.0031, +0.0133]
delta_net	+0.0076 [-0.0008, +0.0159]
sensitivity	+0.6691 [+0.5550, +0.7763]
T (delta_net)	+0.0113

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 14: Recorded belief contrast and sensitivity quantities. Source: software_architecture/sw_arms_v1_s7/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0217 [+0.0115, +0.0333]
machinery	+0.0094 [+0.0038, +0.0152]
delta_net	+0.0122 [+0.0053, +0.0202]
sensitivity	+0.6691 [+0.5550, +0.7763]
T (delta_net)	+0.0183

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 15: Recorded belief contrast and sensitivity quantities. Source: software_architecture/sw_arms_v1_s123/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0148 [+0.0050, +0.0264]
machinery	+0.0182 [+0.0094, +0.0280]
delta_net	-0.0034 [-0.0156, +0.0089]
sensitivity	+0.6691 [+0.5550, +0.7763]
T (delta_net)	-0.0050

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.